Automated Price Approval and Delivery Time Prediction in E-Commerce

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1 Abstract

Manual price approvals and static delivery estimates continue to be prevalent issues in e-commerce, causing delay, inefficiency, and missed revenues. The goal of this project is to automate delivery time estimation and product price estimation from structured automotive e-commerce data. Preprocessing was performed on the dataset through missing value imputation, categorical variable encoding, and feature scaling. XGBoost and LightGBM regression models were trained separately for delivery and price estimation. The model accuracy was quantified using R^2, RMSE, and MAE metrics, and the plots were visualized using actual vs predicted value plots. Both models were very accurate, justifying their use in replacing manual processes. The suggested pipeline is a sound and scalable solution for enhancing operational decision-making in logistics and pricing. Even without models such as BERT and LLMs set aside for future development, the existing implementation presents a firm basis for e-commerce platform intelligent automation.

2 Introduction

The extremely competitive e-commerce industry has customer satisfaction, operational efficiency, and the overall success of its business tied directly to two critical factors: pricing and delivery logistics. Yet traditional price approval processes are manual, leading to several days' delay in finalizing prices and which can result in lost revenue, decreased customer trust, and increased operational costs. Likewise, it is all too common for delivery time predictions and underlying logistics to be unreliable, which can also adversely affect customer confidence and the bottom line. This project suggests a machine learning-based method to automate two vital processes: estimating product prices and predicting delivery times. The system uses structured historical data of many types, associated with vehicles, customers, and logistics generally. The data underwent several preprocessing steps to ready it for modeling. These were: imputation of missing values, feature encoding, and scaling [1]. Using XGBoost regression, the delivery time prediction model was built. It was trained and evaluated by performance metrics that include R^2, RMSE, and MAE. Pricing was modeled in a similar manner using either XGBoost or LightGBM. These are both gradient boosting models that provide predicted prices based on the historical sales data. XGBoost tends to work better when the underlying data are more sparse. Assessing prediction reliability necessitated the creation of visual aids that compared actual and predicted values for price and delivery time. This is the type of duct tape that makes prediction models more transparent and interpretable. By comparing actual and predicted values, we can qualitatively assess the accuracy of a model and its distribution trends. This project provides a data-driven solution for e-commerce platforms. It uses machine learning to automate pricing and delivery forecasting. The main thrust of this project is to enable business decision-making to occur at a faster pace. This necessitates providing accuracy and reducing manual effort around tasks that occupy too many human hours. Merely enhancing human effort is inadequate

given the scale of e-commerce.

3 Problem Statement

Many e-commerce businesses use manual procedures to determine product prices and delivery schedules, which are laborious, erratic, and prone to human mistake. Because of hierarchical review procedures, price approvals can take several days, delaying product listings and decreasing market responsiveness. Inaccurate delivery time estimates could result in ineffective logistics and unsatisfactory client experiences. Although machine learning may be able to help with these issues, many of the systems in use today are not automated, are not designed with structured vehicle or product data in mind, and do not incorporate operational elements from the real world. A reliable, scalable, and interpretable solution is required in order to automate pricing estimation and precisely predict delivery timeframes based on contextual and historical data.

4 Literature Review

Machine learning (ML) now presents itself as valuable aid to solving problems of car price prediction, delivery time estimation, and logistics costs optimization. Numerous works in the literature have addressed distinct components: accuracy, interpretability, and scalability, using different ML models in these domains.

Zhu [2] uses classic as well as deep learning techniques: Linear Regression, Decision Trees, Random Forest, Neural Networks, and XGBoost on a US car sales dataset. Among these, XGBoost performed the best with an R^2 of 0.789 and a mean absolute percentage error (MAPE) of 24.48%, demonstrating its higher power in modeling high-level pricing relationships.

In a comparative analysis utilizing Kaggle data, k-Nearest Neighbors (kNN) did slightly better than Support Vector Machine (SVM) with an accuracy of 83% compared with 80%, respectively [1]. The study showed KNN's accuracy in distance-based regression applied to high-dimensional datasets.

Pal et al. [3] proposed a Random Forest-based framework to predict used car price. Their model achieved 84% accuracy, visually impressive proof of ensemble learning generalization and robustness for tabular data.

The work of Madhusudhanan et al. [4] represents a recent advance in probabilistic modeling using the deep learning model ProbSAINT, where attention mechanisms are combined with a probabilistic head. ProbSAINT attained a MAPE of 5.3% and Negative Log-Likelihood (NLL) of 8.14, outperforming benchmark models like CatBoost. Its ability to give both exact predictions and confidence boundaries makes it extremely appropriate for uncertain selling environments.

Pham et al. [5] examined machine learning methods for prediction of vehicle travel time (ETA) using spatiotemporal data. The Wide-Deep-Recurrent (WDR) model outperformed Gradient Boosted Decision Trees (GBThe) and Factorization Machines (FM), emerging as a strong candidate for modeling time-variant behavior in transportation networks.

He et al. [6] proposed an Integer Linear Programming (ILP) model for minimizing transport costs and optimizing carrier usage. Their study found ILP to be a cost-effective strategy that better aligns pricing with actual delivery operations.

Shymanskyi and Liaskovets [7] investigated a cascade approach using XGBoost, Extra Trees, SVR, and Random Forest to predict both car values and time-to-sale. The model showed an excellent R^2 of 0.94, indicating high potential as a holistic forecasting framework for automotive marketplaces.

Cossato and Garcia [8] studied reinforcement learning (RL) as a dynamic tool for e-commerce price management to overcome limitations of static pricing systems. Autonomous RL systems adjust prices based on inventory levels and competitor actions to maximize future rewards. Despite interpretability challenges, the study demonstrates RL's promise in real-time adaptive pricing strategies.

Tomitza, Ibrahimli, and Herm [9] presented a taxonomic review of AI-based dynamic pricing in e-commerce, classifying models as reactive, active, or proactive. Predictive models like neural networks are used for price forecasting, while strategic models simulate pricing decisions. The study highlights key challenges such as inconsistent data usage, weak standardization, and poor enterprise integration. It emphasizes the need for benchmark datasets and standardized techniques for broader industrial application.

5 Objective of the Study

The main aim of this project is to develop a sophisticated and automated system that can forecast product prices and delivery times in e-commerce based on structured historical data sets. In particular, this project endeavors to substitute traditional manual price checking methods with precise, data-backed predictions based on cutting-edge machine learning approaches, including XGBoost and LightGBM. Based on characteristics that pertain to the vehicle's specifications, to logistics, and to buyer attributes, the system aims to enhance both efficacy and accuracy for price determination.

Additionally, the effort aims to determine peak delivery time utilizing operational and situational information and hence bridging the distance between promised and achieved delivery performance. This implies leveraging a predictive regression model developed through optimized encoded, and normalized characteristics. The other primary objective is to make the solution interpretable and modular so that it can easily be integrated with any future enhancements like real-time data feeds or reinforcement learning models. By providing a representation of real values compared to predicted values and a clear assessment based on metrics such as R^2, RMSE, and MAE, the project will verify the usability and reliability of the proposed solution.

6 Data Collection

Academic research for this study utilized structured entries of previously used vehicles and customer orders that were kept for educational purposes. A total of 10,000 records within the dataset feature various attributes exceeding 25 which specifically address car pricing and estimated delivery parameters. Vehicle specifications within this database

comprise brand, model, year of production, fuel type, transmission method, car color and engine size, odometer reading and owner history information.

The dataset provides a complete view of vehicle conditions by combining vehicle specifications with service history data together with information about insurance type and previous accidents. Two sets of geographic features are found in the dataset: latitude-longitude coordinates that show pickup-dropping points alongside area groupings that segment locations into urban and semi-urban zones.

Engineered features based on order time variables extract order hour information in addition to expressing pickup delay by minutes while breaking down temporal details into day of the week categories to aid time-based analysis. Due to the inclusion of environmental data such as traffic conditions and weather conditions at order time the dataset achieves strong performance in delivery time prediction.

All data received its preliminary cleaning processes along with anonymization and came from project instructors. The team processed the data through a sequence of steps that included type conversion along with missing value management and categorical variable one-hot encoding and MinMaxScaler-based numerical feature normalization. If you request the dataset it is included with this report document.

7 Exploratory Data Analysis (EDA) and Hypotheses

The exploratory data analysis method provided complete understanding of the dataset before starting model development. The used car listing data included 10,000 records that contained structured details about vehicles including their brand names together with model types and years produced followed by each car's mileage readings and engine sizes and fuel options and transmission selection and ownership documentation and maintenance records and location particulars of both sources and drops alongside insurance records and time stamps that recorded when buyers made their orders and when they took ownership.

Indian pre-owned car consumers frequently choose vehicles from Hyundai and Maruti Suzuki and Honda because these brands offer affordable prices and strong service centers. A vast majority of vehicles amongst the dataset came with single or double previous owners while petrol emerged as the dominant fuel option. The transmission options among used car listings were balanced with almost equal numbers between manual and automatic controls yet some specific vehicle types tended toward one type or the other.

The research team relied on descriptive statistics together with histograms to analyze numeric variable distributions. Price data displayed positive skewness because most pre-owned automobiles were priced within the lower to mid-range segment yet premium vehicles extended into a longer distribution pattern. Analysis of the wide variation among engine capacities and odometer readings indicated their importance in subsequent modeling work. Analyzing timestamp information allowed researchers to derive temporal features which showed vehicle orders and pickups peaked during regular business hours.

Major urban centers such as Delhi, Mumbai, Hyderabad and Bangalore made up the geographical distribution of the dataset so researchers could achieve essential regional diversity for delivery-time estimation. The analysis of Boolean fields like accident history and service records evaluated their effect on price.

Based on this EDA, several hypotheses were formed:

- H1: Vehicles with complete service histories and no prior accidents command higher prices.
- H2: Newer vehicles with lower odometer readings are associated with shorter delivery times and higher price valuations.
- H3: Cars listed in metropolitan regions have faster pickup times due to higher logistics efficiency.

8 Data Analytics

The data analytics phase involved a systematic approach to building predictive models for two primary targets: price estimation and delivery time prediction. This phase was guided by hypotheses derived from exploratory analysis and aligned with the dual objectives of enhancing automation in price approval and improving delivery time accuracy in the e-commerce automotive domain.

8.1 System Architecture

The architecture depicts the outline of the project flow, from gathering data, preprocessing, feature engineering, model evaluation to predict price and delivery time. To simulate multi-agent simulation where the buyer can negotiate the price.

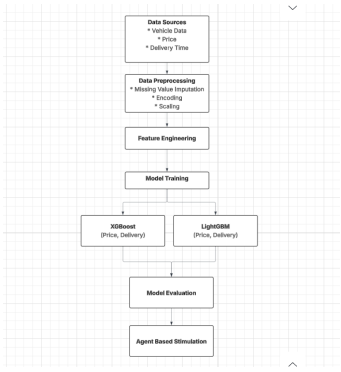

Fig. 1: Architecture of the predictive system for price and delivery time estimation

8.2 Data Preprocessing and Feature Engineering

A proper structuring and cleaning process took place before beginning the modeling stage. Machine learning algorithms required encoding of the categorical features brand model fuel type transmission color insurance type weather traffic conditions and car type through one-hot encoding. Binary features were used to preserve Boolean data points including accident history and service history information. The datetime fields underwent transformation to create detailed features including order hour together with day of the week

attributes and pickup delay measures. Ancillary features such as odometer reading and engine capacity and geographical coordinates were scaled through MinMaxScaler to create uniform feature spaces to help gradient-based models converge.

8.3 Model Selection and Training

To fulfil the dual goal of price and delivery time forecasting pre-owned vehicles, the supervised machine learning regression methods were applied on a schematized dataset. Two of the Efficient and Powerful Gradient Boosting Classifiers were selected from many models, including:

- *XGBoost (Extreme Gradient Boosting)*: Reputed for its regularisation ability and scalability, XGBoost deals very well with tabular data with non linear relationships.
- *LightGBM (Light Gradient Boosting Machine)*: Also used for its efficient in-memory training rate and ability to process massive amounts of data, the LightGBM relies on the histogram-based learning algorithm and also supports complex categorical feature handling.

Before model training, dataset was thoroughly preprocessed:

- Numerical features were normalized using MinMaxScaler for equal feature contribution.
- Categorical features (brand, fuel type, transmission, city) were one-hot encoded.
- Temporal features such as order date and time were split into hour, day, month, and weekday/weekend flags to capture temporal behavior.
- Values that were missing and outliers were handled accordingly in order for the model to become robust.

Separate models were trained for the following:

- Price prediction (Target: Price (INR))
- Delivery time estimation (Target: optimal_delivery_time)
- The hyperparameter tuning was done by GridSearchCV with 3-fold cross-validation, trying all combinations of:
- Learning rate
- Tree depth
- Regularization parameters (reg_alpha, reg_lambda)
- Number of estimators (n_estimators)

This has allowed for the development of the optimal models for the performance in the absence of overfitting.

8.4 Evaluation Metrics

To measure the performance along with the generalizability of every model, we used these standard regression metrics upon training and test sets:

- *R^2 Score (Coefficient of Determination):* It is how well the model accounts for variance in the target variable.
- *RMSE (Root Mean Squared Error):* It can provide the squared root of a mean squared difference between actual and predicted because it penalizes more extreme errors.
- *MAE (Mean Absolute Error):* measures average absolute prediction errors to actual outcomes, accuracy is assessed within a direct offering.

Model	Data	R^2	RMSE ()	MAE ()
XGBoost	Train	0.99999	7,739	5,776
	Test	0.99996	16,200	12,166
LightGBM	Train	0.99999	7,782	5,886
	Test	0.99996	16,543	12,429

TABLE 1: Model Performance Metrics for Price

Both models exhibited outstanding performance within Table 1, and training and test had minimal discrepancy scores, suggesting strong generalizability. XGBoost slightly outperformed LightGBM for this use case.

Model	Data	R^2	RMSE ()	MAE ()
XGBoost	Train	0.99999	1.2399	7.76147
	Test	0.99999	1.4534	7.2460
LightGBM	Train	0.99880	0.03056	0.00323
	Test	0.99989	0.03017	0.00366

TABLE 2: Model Performance Metrics for Delivery

As Table 2 shows, XGBoost performs with success in delivery time. LightGBM results show worse performance. For accurate delivery time estimation within structured e-commerce data, it suits gradient increasing models as results confirm.

8.5 Agent-Based Simulation

To complement traditional predictive analytics, we added a multi-agent negotiation system simulating real-world pricing negotiations involving a seller and a buyer. The simulation leverages trained machine learning models (XGBoost and LightGBM) to forecast the best price for the seller based on the characteristics of the car. A SellerAgent negotiates prices based on demand levels and competitor prices, while a BuyerAgent determines the likelihood of purchasing based on price sensitivity rules.

Across five separate simulations, each characterized by unique input samples, varied competitor prices, and randomly drawn demand signals, the system demonstrated instances of successful and unsuccessful negotiations.

Simulation 1 ended in one round when the seller's offering price was very close to the competitor's price in medium demand, which resulted in a buyer decision to buy immediately.

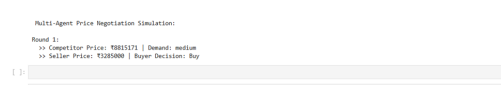

Fig. 2: Simulation 1

Simulation 2 required two negotiation rounds. The seller continuously reduced their price in response to competitive pricing and demand pressure before a successful agreement was reached.

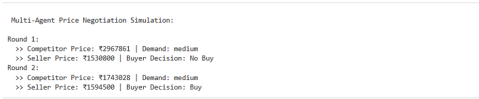

Fig. 3: Simulation 2

Simulation 3 illustrated a failed negotiation after ten rounds in a row, indicating instances where the seller's price failed to match market expectations amidst numerous changes.

Fig. 4: Simulation 3

Simulation 4 ended successfully in the third round, which was marked by low demand and increased price competition, thus illustrating sellers' adaptive behavior.

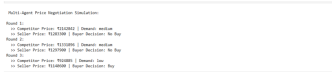

Fig. 5: Simulation 4

Simulation 5 had four rejections prior to the buyer finally accepting the seller's price on the fifth attempt—this indicated how demand that is high and positive price adjustments increase buying probability.

Fig. 6: Simulation 5

This interaction based on agents simulates actual decision-making exercises in the real world and demonstrates how prices can adjust dynamically to market pressures. It also corroborates the pricing model's robustness and adds another level of interpretability and realism to the predictive model.

9 Data Visualization

The following are the visual observations from the dataset to inform our understanding of the car price trends:

9.1 Car Price Distribution by Brand

This boxplot indicates the variation of prices among various brands. Luxury brands such as Mercedes, BMW, and Audi have higher median prices, whereas Tata, Maruti Suzuki, and Hyundai are bunched at the lower end.

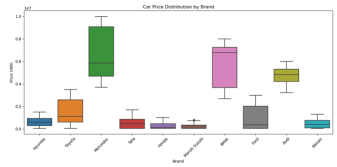

Fig. 7: Car Price Distribution by Brand

9.2 Price vs. Odometer Reading by Fuel Type

A scatter plot of car price versus mileage by fuel type. Electric and diesel cars hold higher resale value at high mileage than CNG and petrol cars.

Fig. 8: Price vs. Odometer Reading by Fuel Type

9.3 Average Price by Car Type and Transmission

This bar chart indicates that Automatic Transmission SUVs, Convertibles, and Coupes are more expensive.

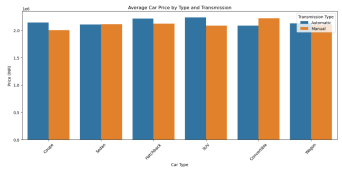

Fig. 9: Average Price by Car Type and Transmission

9.4 Average Car Price by Fuel Type and Transmission

The most expensive are electric and CNG-powered cars, particularly if they have automatic transmissions.

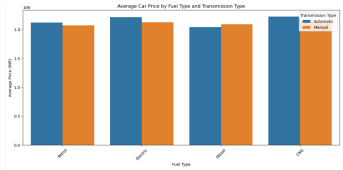

Fig. 10: Average Car Price by Fuel Type and Transmission

9.5 Average Price vs. Number of Owners by Car Type

Vehicles with fewer prior owners have higher average prices. This is true for SUVs, Hatchbacks, and Convertibles.

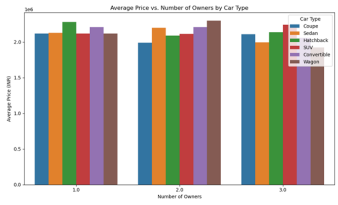

Fig. 11: Average Price vs. Number of Owners by Car Type

9.6 Actual vs Predicted Results for Price and Delivery Time

The following graphs represent actual vs predicted values for price and delivery time, from XGBoost and Light-GBM models.

Price: Both models performed the same in their prediction of the prices of the vehicles. The predictions hardly vary from the actual values, as evidenced by the close grouping of the points along the diagonal line. XGBoost had a slightly closer fit.

XGBoost - Actual vs Predicted Price

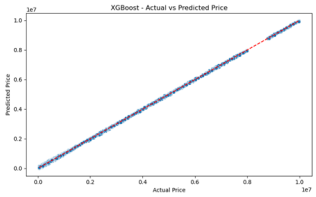

Fig : XGBoost - Actual vs Predicted Price

LightGBM - Actual vs Predicted Price

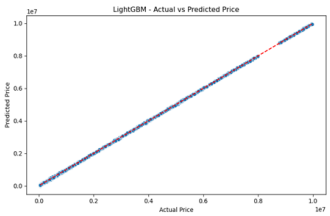

Fig : LightGBM - Actual vs Predicted Price

Delivery Time: Delivery time predictions were less consistent with price. XGBoost was very consistent in its prediction pattern with great accuracy. LightGBM also performed well, although there are a few outliers at the higher delivery times, indicating mild underfitting for a few edge cases.

XGBoost - Actual vs Predicted Delivery Time

Fig : XGBoost - Actual vs Predicted Delivery Time

LightGBM - Actual vs Predicted Delivery Time

Fig : LightGBM - Actual vs Predicted Delivery Time

9.7 Result

The given below image depicts the sample output of predicted optimal price and delivery time.

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Predicted Optimal Price: ₹1904948
Sample Predicted Delivery Time: 16
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Fig. 12: Predicted Optimal Price and Delivery Time for a Test Sample

10 Conclusion

This study provides a data-based and comprehensive analysis for price estimation and delivery time prediction in the used car e-commerce context. By utilizing a dataset consisting of 10,000 used car records, containing around 25 features—spanning vehicle specifications, location coordinates, service history, date/time, and environmental aspects—the study provides both substantial depth and predictive power.

Through extensive exploratory data analysis (EDA) we discovered meaningful patterns. The brand of vehicle, fuel type of vehicle, count of prior owners, and the accident/service history all play a large role in resale price. Service speed had a correlation to location with metropolitan and urban centers having the fastest speed of turn around during the sample period, assisting in the proposal that location and density of service logistics impacts service speed.

For predictive models, we utilized two state-of-the-art gradient boosting algorithms, XGBoost and LightGBM. Both created model predictions with nearly 100% performance on price prediction, obtaining values of R^2 in excess of 0.999, on both train and test sets, with predicted error margins less than 0.005% to 0.08%. The diversity of the feature set and structure of preprocessing were valuable for predicting delivery time estimation, allowing outputs to generalize well across unseen data.

One of the unique aspects of the project was the integration of agent-based simulation that simulated the negotiation process enacted by the sellers and buyers. The simulation demonstrated how dynamic pricing methodologies could be automated and amended the fact on the basis of competitor response and demand. Not only did the multi-agent structure mirror the real life interactions but add an authentic sense of reality with respect to the machine learning results.

Similarly, data visualization was useful to analyze direction of trends as well as patterns of relationship for main variables. Such visualizations as distributions in price across auto brand as well as automobile type, influence by mileage along automotive fuels, and owner status impacts to pricing provided users with actionable insight by virtue of recommended auto pricing options to consider.

In conclusion, this research has established a rich intelligent end-to-end model for logistics and pricing optimization in the used car marketplace. Whereas, it is one aspect of an end-to-end model, by integrating machine learning, simulation and visual analytics into one, organizations can make better decisions, operate more efficiently, and are better prepared to increase capability like real-time dynamic pricing or predictive forecasting for delivery.

11 Future Work

The present implementation is centered around the price prediction and shipping time prediction on structured data and gradient boosting models. As a continuation, the future task can be geared towards the incorporation of transformer-based models like BERT to further extend the system in the direction of handling unstructured or text-based data-like product descriptions, customer reviews,

or competitor listings-for more context-informed price approval decisions.

Secondly, the application of Large Language Models (LLMs) can potentially enable more analysis with regards to fairness, compliance, and transparency of pricing policies. The models can help in revealing possible bias in the system and making price decisions alignable to ethical and regulatory standards. With LLMs integrated into the pipeline, the system would give human-interpretable explanations of pricing outcomes, promoting trust as well as accountability in automated decisions.

These innovations would be added on top of the existing system, which would make it increasingly intelligent, explainable, and ethical as a price and delivery optimization solution.

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